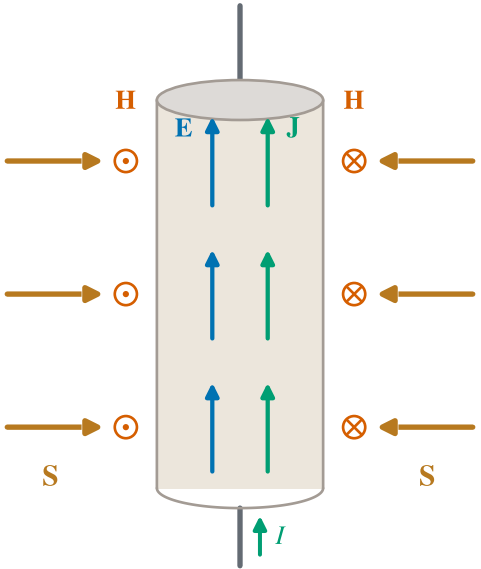
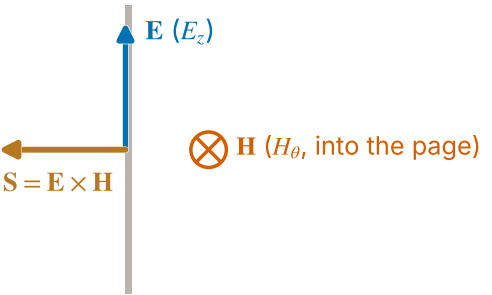


inflow per unit length: $2\pi a\, E_z H_\theta = E_z I = I^2 R/\ell$

$$H_\theta = \frac{I}{2\pi a}$$ (Ampère)



the inflow equals the ohmic heat inside